Note 6: QEC2

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6.1 Summary of Progress

1. Reviewed the fundamental theory of stabilizer codes, derived the stabilizer projector and verified the correspondence between abelian Pauli subgroups and symplectic selforthogonal binary codes.
2. Rigorously constructed CSS codes from classical linear codes, proved the stabilizer representation of CSS codes, and clarified the linear independence of cosetsummed codeword states.
3. Completed the full errorcorrection workflow for CSS codes with mathematical proofs, and initiated the categorification framework for both classical and quantum errorcorrection models.

We have the following relationships about various quantum code

$$\left\{ \begin{array}{c} \text{surface} \\ \text{code} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{topological} \\ \text{code} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{CSS} \\ \text{code} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{stabilizer} \\ \text{code} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{Quantum} \\ \text{code} \end{array} \right\}$$

Definition 1 (Quantum Code) If an isometry $\mathcal{U}_E$ can be expressed as: a unitary matrix U_E acts on k logical qubits, together with $n-k$ ancilla qubits initialized in the $|0\rangle$ state:

$$\mathcal{U}_E|\psi\rangle = U_E\big(|\psi\rangle \otimes |0\rangle^{\otimes n-k}\big)$$

- A **quantum code** (or **codespace**) C is the image of an encoding isometry $\mathcal{U}_E$:

$$C = \mathrm{Im}(\mathcal{U}_E)$$

- Any element in $\mathcal{C}$ is called a **codeword**.
- Any circuit implementation of U_E is called an **encoding circuit**.
- We use the bar notation $|\bar{\cdot}\rangle$ to denote an **encoded state**, for example:

$$|\overline{0}\rangle = |000\rangle$$

6.2 Stabilizer code

6.2.1 Projector to Stabilizer Code

Definition 2 (Stabilizer Code Model) As a subclass of the general quantum error correction model:

1. Equipped with an additional **abelian subgroup** $\mathcal{G}$ of the Pauli group $\mathcal{P}$ on $\mathcal{H}_{\mathrm{P}}$, such that $-\mathbb{I} \notin \mathcal{G}$.
2. The code subspace is defined by

$$C_{\mathcal{G}} = \big\{|\psi\rangle \in \mathcal{H}_{\mathrm{P}} \,\big|\, g|\psi\rangle = |\psi\rangle,\ \forall g \in \mathcal{G}\big\}.$$

3. The noise operation $\mathcal{E}$ is restricted to channels whose Kraus operators belong to $\mathcal{P}$.
4. The recovery operation $\mathcal{R}$ is restricted to coset-based recovery induced by $\mathcal{P}/\mathcal{G}$.

Theorem 3 *Let*

$$P_{\mathcal{G}} := \frac{1}{|\mathcal{G}|}\sum_{g\in\mathcal{G}} g$$

then we have $P_{\mathcal{G}}$ is a projector to $C_{\mathcal{G}} = \mathrm{Im} P_{\mathcal{G}}$.

Proof Since $R_g : \mathcal{G} \to \mathcal{G}, h \mapsto hg$ is bijective, so

$$P_{\mathcal{G}} g = \frac{1}{|\mathcal{G}|}\sum_{h\in\mathcal{G}} hg = \frac{1}{|\mathcal{G}|}\sum_{h\in\mathcal{G}} h = P_{\mathcal{G}},$$

then

$$P_{\mathcal{G}}{}^2 = \frac{1}{|\mathcal{G}|}\sum_{h\in\mathcal{G}} P_{\mathcal{G}} h = \frac{1}{|\mathcal{G}|}\sum_{h\in\mathcal{G}} P_{\mathcal{G}} = P_{\mathcal{G}}$$

So $P_{\mathcal{G}}$ is a projector.

For $|\psi\rangle \in \mathrm{Im} P_{\mathcal{G}} \Rightarrow |\psi\rangle = P_{\mathcal{G}}|\phi\rangle$, then for all $g \in \mathcal{G}$ we have

$$g|\psi\rangle = gP_{\mathcal{G}}|\phi\rangle = P_{\mathcal{G}} g|\phi\rangle = P_{\mathcal{G}}|\phi\rangle = |\psi\rangle$$

So $|\psi\rangle \in C_{\mathcal{G}}$. If for all $g \in \mathcal{G}$, $g|\psi\rangle = |\psi\rangle$, then

$$P_{\mathcal{G}}|\psi\rangle = \frac{1}{|\mathcal{G}|}\sum_{g\in\mathcal{G}} g|\psi\rangle = \frac{1}{|\mathcal{G}|}\sum_{g\in\mathcal{G}} |\psi\rangle = |\psi\rangle \in \mathrm{Im} P_{\mathcal{G}}$$

In all, we have $C_{\mathcal{G}} = \mathrm{Im} P_{\mathcal{G}}$. ■

Theorem 4 *Given a positive integer n, the composition map*

$$\Psi : \mathcal{P}_n \xrightarrow{\pi} \mathcal{P}_n/\langle iI\rangle \xrightarrow{\Phi} \mathbb{F}_2^{2n}$$

induces a surjective map

$$\left\{G \leq \mathcal{P}_n \,\middle|\, G \text{ is abelian and } -I \notin G\right\} \xrightarrow{\widetilde{\Psi}} \left\{C \leq \mathbb{F}_2^{2n} \,\middle|\, C \text{ is symplectic self-orthogonal}\right\}.$$

Furthermore, if $C = \mathbb{F}_2 c_1 \oplus \cdots \oplus \mathbb{F}_2 c_k$,then there exist $g_j \in \Psi^{-1}(c_j)$ for each j, such that

$$G = \langle g_1, \ldots, g_k\rangle$$

and $\widetilde{\Psi}(G) = C$.

In weeknote 5 , we have proved this theorem. Using the same notation as in the previous theorem, we have

Theorem 5

$$P_{\mathcal{G}} = \prod_{j=1}^{k} \frac{I + g_j}{2}$$

Proof

$$\begin{aligned} P_{\mathcal{G}} :&= \frac{1}{|\mathcal{G}|}\sum_{g\in\mathcal{G}} g = \frac{1}{|\mathcal{G}|}\sum_{(\epsilon_1,\ldots,\epsilon_k)\in\mathbb{F}_2^k} g_1^{\epsilon_1} \ldots g_k^{\epsilon_k} \\ &= \frac{1}{|\mathcal{G}|}\prod_{j=1}^{k}(I + g_j) = \prod_{j=1}^{k}\frac{I + g_j}{2} \end{aligned}$$

■

6.2.2 Code distance and detection capacity of stabilizer codes

Definition 6 (Quantum code distance of stabilizer codes.) Let $\mathcal{G} \leq \mathcal{P}_n$ be the stabilizer group. The **code distance** d is

$$d = \min \left\{ \operatorname{wt}(P) \,\middle|\, P \in C_{\mathcal{P}_n}(\mathcal{G}) \setminus \mathcal{G} \right\},$$

where $C_{\mathcal{P}_n}(\mathcal{G})$ denotes the centralizer of $\mathcal{G}$ in $\mathcal{P}_n$, and $\operatorname{wt}(P)$ is the Pauli weight (number of nonidentity tensor factors).

Remark 7 Let $\Psi \colon \mathcal{P}_n \to \mathbb{F}_2^{2n}$, then $\operatorname{wt}(P)$ coincides with the *symplectic weight* of $\Psi(P)$, that is

$$\operatorname{wt}(P) = \operatorname{wt}_s(\Psi(P)) = \#\big\{k \mid (\Psi(P)_k, \Psi(P)_{k+n}) \neq (0,0)\big\}.$$

Recall that

Theorem 8 *The quantum operation $\mathcal{E}(\rho) = \sum_i E_i \rho E_i^\dagger$ exists a recovery operation $\mathcal{R}$ if and only if there exist scalars $\alpha_{ij} \in \mathbb{C}$ such that*

$$P E_i^\dagger E_j P = \alpha_{ij} P.$$

Theorem 9 *Let $\mathcal{G}$ be the stabilizer group of a stabilizer code with distance d, and let*

$$t = \left\lfloor \frac{d-1}{2} \right\rfloor.$$

For any error channel

$$\mathcal{E}(\rho) = \sum_i E_i \rho E_i^\dagger$$

with each $E_i \in \mathcal{P}_n$ and $\operatorname{wt}(E_i) \leq t$, then there exists a recovery operation.

Proof Let $t = \left\lfloor \frac{d-1}{2} \right\rfloor$. Take any set of Pauli errors $\{E_i\}$ with $\operatorname{wt}(E_i) \leq t$. For any pair E_i, E_j, we have

$$\operatorname{wt}(E_i^\dagger E_j) \leq \operatorname{wt}(E_i) + \operatorname{wt}(E_j) \leq 2t \leq d-1.$$

By definition of code distance, any Pauli operator of weight $< d$ is either in $\mathcal{G}$ or not in $C_{\mathcal{P}_n}(\mathcal{G})$.

- If $E_i^\dagger E_j = g \in \mathcal{G}$, then $P E_i^\dagger E_j P = PgP = PP = P$.
- If $E_i^\dagger E_j \notin C_{\mathcal{P}_n}(\mathcal{G})$, there exists $g \in \mathcal{G}$ such that $E_i^\dagger E_j g = -g E_i^\dagger E_j$.
- Since $g \in \mathcal{G}$, we have $gP = P$ and $Pg = P$.

$$\begin{aligned} P E_i^\dagger E_j P &= P E_i^\dagger E_j (gP) \\ &= P\big(E_i^\dagger E_j g\big) P \\ &= P\big(-g E_i^\dagger E_j\big) P \\ &= -(Pg) E_i^\dagger E_j P \\ &= -P E_i^\dagger E_j P. \end{aligned}$$

Rearranging gives $2 P E_i^\dagger E_j P = 0$, hence $P E_i^\dagger E_j P = 0$.

In both cases, $P E_i^\dagger E_j P = \alpha_{ij} P$ for some scalar $\alpha_{ij} \in \mathbb{C}$. By the KnillLaflamme condition, a recovery operation $\mathcal{R}$ exists. Thus the code corrects all errors of weight $\leq t$. ■

6.3 CSS code

Definition 10 Let $V_1, V_2 \subseteq \mathbb{F}_2^n$ be linear subspaces satisfying $V_1 \perp V_2$. The CSS code is the stabilizer code with stabilizer group

$$\mathcal{G} = \left\{ X^a Z^b \,\middle|\, (a,b) \in V_1 \times V_2 \subseteq \mathbb{F}_2^{2n} \right\}.$$

Remark 11 Since $V_1 \perp V_2$, for $(a_1, b_1), (a_2, b_2) \in V_1 \times V_2$, we have

$$a_1 \cdot b_2 + a_2 \cdot b_1 = 0,$$

so $V_1 \times V_2$ is symplectic self-orthogonal

Remark 12 In the textbook [1], the auther use the notation $\mathrm{CSS}(C_1, C_2)$ is the CSS code with $C_2 = V_1 \subseteq C_1 = V_2^\perp$. In other words, the stabilizer group is

$$\mathcal{G} = \left\{ X^a Z^b \,\middle|\, (a,b) \in V_1 \times V_2 \subseteq \mathbb{F}_2^{2n} \right\} = \left\{ X^a Z^b \,\middle|\, (a,b) \in C_2 \times C_1^\perp \subseteq \mathbb{F}_2^{2n} \right\}.$$

Lemma 13 *Let $H^{\otimes n} = \underbrace{H \otimes \ldots \otimes H}_{n}$ be tensor product of Hadmard tranformation, and $w \in \mathbb{F}_2^n$, then*

$$H^{\otimes n} |w\rangle = \frac{1}{\sqrt{2^n}} \sum_{\epsilon \in \mathbb{F}_2^n} (-1)^{w \cdot \epsilon} |\epsilon\rangle.$$

Proof Recall that

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

So we have

$$\begin{aligned} H^{\otimes n}|w\rangle = H^{\otimes n}|w_1 \ldots w_n\rangle &= \bigotimes_{j=1}^{n} H|w_j\rangle \\ &= \bigotimes_{j=1}^{n} \frac{|0\rangle + (-1)^{w_j} |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2^n}} \bigotimes_{j=1}^{n} (|0\rangle + (-1)^{w_j} |1\rangle) \\ &= \frac{1}{\sqrt{2^n}} \sum_{(\epsilon_1, \ldots, \epsilon_n) \in \mathbb{F}_2^n} \prod_{j=1}^{n} (-1)^{w_j \epsilon_j} |\epsilon_1, \ldots, \epsilon_n\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_{(\epsilon_1, \ldots, \epsilon_n) \in \mathbb{F}_2^n} (-1)^{w_1 \epsilon_1 + \ldots + w_n \epsilon_n} |\epsilon_1, \ldots, \epsilon_n\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_{\epsilon \in \mathbb{F}_2^n} (-1)^{w \cdot \epsilon} |\epsilon\rangle. \end{aligned}$$

■

Lemma 14 *Let C be a linear code. If $x \in C^\perp$ then $\sum_{y \in C} (-1)^{x \cdot y} = |C|$, while if $x \notin C^\perp$ then $\sum_{y \in C} (-1)^{x \cdot y} = 0$.*

Proof If $x \in C^\perp$ then

$$\sum_{y \in C} (-1)^{x \cdot y} = \sum_{y \in C} (-1)^0 = |C|.$$

If $x \notin C^{\perp}$, then

$$T_x : C \to \mathbb{F}_2, y \mapsto x \cdot y$$

is a linear function with $T_x \neq 0$, so

$$\dim \ker T_x = \dim C - \dim \mathbb{F}_2 = \dim C - 1,$$

thus

$$\#\{y \in C : x \cdot y = 0\} = \#\ker T_x = \frac{|C|}{2} = \#\{y \in C : x \cdot y = 1\}$$

which implies $\sum_{y \in C} (-1)^{x \cdot y} = 0.$ ■

Lemma 15 *For $a, b \in \mathbb{F}_2^n$, denote $X^a = X^{a_1} \otimes ... \otimes X^{a_n}$, $Z^b := Z^{b_1} \otimes ... \otimes Z^{b_n}$, then we have*

$$X^a |x\rangle = |x + a\rangle, \quad Z^b |y\rangle = (-1)^{b \cdot y} |y\rangle.$$

Proof For $(X|0\rangle, X|1\rangle) = (|1\rangle, |0\rangle), (Z|0\rangle, Z|1\rangle) = (|0\rangle, -|1\rangle)$, we have

$$\begin{aligned}
X^a |x\rangle &= X^{a_1} \otimes ... \otimes X^{a_n} |x_1\rangle \otimes ... \otimes |x_n\rangle \\
&= X^{a_1} |x_1\rangle \otimes ... \otimes X^{a_n} |x_n\rangle \\
&= |x_1 + a_1\rangle \otimes ... \otimes |x_n + a_n\rangle = |x + a\rangle \\
Z^b |y\rangle &= Z^{b_1} \otimes ... \otimes Z^{b_n} |y_1\rangle \otimes ... \otimes |y_n\rangle \\
&= Z^{b_1} |y_1\rangle \otimes ... \otimes Z^{b_n} |y_n\rangle \\
&= (-1)^{a_1 \cdot y_1} |y_1\rangle \otimes ... \otimes (-1)^{a_n \cdot y_n} |y_n\rangle = (-1)^{b \cdot y} |y\rangle
\end{aligned}$$

■

Proposition 16 *The **CSS code** $\mathrm{CSS}(C_1, C_2)$ is the quantum code spanned by the coset states*

$$|x + C_2\rangle = \frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} |x + y\rangle, \quad \forall\, x \in C_1.$$

Equivalently,

$$\mathrm{CSS}(C_1, C_2) := \mathrm{span}_{\mathbb{C}} \left\{ |x + C_2\rangle : x \in C_1 \right\}.$$

Proof Recall that the stabilizer group is

$$\mathcal{G} = \left\{ X^a Z^b \,\middle|\, (a, b) \in C_2 \times C_1^{\perp} \subseteq \mathbb{F}_2^{2n} \right\}.$$

Step1: By lemma before

$$P_{\mathcal{G}} = \frac{1}{|\mathcal{G}|} \sum_{g \in S} g = \frac{1}{|\mathcal{G}|} \sum_{(a,b) \in C_2 \times C_1^{\perp}} X^a Z^b = \frac{1}{|C_1^{\perp}||C_2|} \sum_{b \in C_1^{\perp}} Z^b \sum_{a \in C_2} X^a$$

Step2: For all $x \in \mathbb{F}_2^n$,

$$\begin{aligned}
P_{\mathcal{G}} |x\rangle &= \frac{1}{|C_1^{\perp}||C_2|} \sum_{b \in C_1^{\perp}} Z^b \sum_{a \in C_2} X^a |x\rangle = \frac{1}{|C_1^{\perp}||C_2|} \sum_{b \in C_1^{\perp}} Z^b \sum_{a \in C_2} |x + a\rangle \\
&= \frac{1}{|C_1^{\perp}||C_2|} \sum_{b \in C_1^{\perp}} \sum_{a \in C_2} (-1)^{b \cdot (x + a)} |x + a\rangle,
\end{aligned}$$

Since $b \in C_2 \subseteq C_1$, we have $a \cdot b = 0$. Then we use the lemma

$$\begin{aligned} P_{\mathcal{G}}|x\rangle &= \frac{1}{|C_1^\perp||C_2|}\sum_{a\in C_2}\left(\sum_{b\in C_1^\perp}(-1)^{b\cdot x}\right)|x+a\rangle \\ &= \begin{cases} 0 & \text{if } x \notin \left(C_1^\perp\right)^\perp = C_1 \\ \frac{1}{|C_1^\perp||C_2|}\sum_{a\in C_2}|C_1^\perp||x+a\rangle & \text{if } x \in C_1 \end{cases} \\ &= \begin{cases} 0 & \text{if } x \notin C_1 \\ \frac{1}{|C_2|}\sum_{a\in C_2}|x+a\rangle & \text{if } x \in C_1 \end{cases} \end{aligned}$$

Thus $C_{\mathcal{G}} = \operatorname{Im} P_{\mathcal{G}} = \operatorname{CSS}(C_1, C_2)$. ■

Remark 17 The dimension is 2^k where $k = \dim C_1/C_2 = \dim C_1 - \dim C_2$. It encodes k logical qubits into n physical qubits.

Remark 18 If $x + C_2 \neq x' + C_2$, then for all $y, y' \in C_2$ does $x + y \neq x' + y' \implies |x+y\rangle$ is orthonormal with $|x' + y'\rangle$, and therefore $|x + C_2\rangle$ and $|x' + C_2\rangle$ are orthonormal states.

Error (noise) for CSS code

We encode $|x + C_2\rangle$

$$|\psi\rangle = |x + C_2\rangle = \frac{1}{\sqrt{|C_2|}}\sum_{y\in C_2}|x+y\rangle$$

The state $|x + C_2\rangle$ is affected by noise $\mathcal{E}$, then the corrupted state is

$$|\psi_0\rangle = \mathcal{E}|\psi\rangle = \frac{1}{\sqrt{|C_2|}}\sum_{y\in C_2}(-1)^{(x+y)\cdot e_2}|x+y+e_1\rangle, \quad e_1, e_2 \in \mathbb{F}_2^n$$

Recover operation for CSS code

The recover process can be expressed by the chain

$$|\psi_0\rangle \mapsto |\psi_0\rangle\otimes|\vec{0}\rangle \xmapsto{U_{H1}} |\psi_1\rangle \xmapsto{\mathrm{tr}_{\mathcal{H}}} |\psi_2\rangle \xmapsto{U_{e1}} |\psi_3\rangle \xmapsto{H^{\otimes n}} |\psi_4\rangle \mapsto |\psi_4\rangle\otimes|\vec{0}\rangle \xmapsto{U_{H2}} |\psi_5\rangle \xmapsto{\mathrm{tr}_{\mathcal{H}}} |\psi_6\rangle \xmapsto{U_{e2}} |\psi_7\rangle \xmapsto{H^{\otimes n}} |\psi\rangle$$

Step 1: Introducing an ancilla containing sufficient qubits, and initially in the all zero state $|\vec{0}\rangle$.

$$|\psi_0\rangle \mapsto |\psi_0\rangle \otimes |\vec{0}\rangle$$

Using reversible computation to apply the parity matrix H_1 for the code C_1:

$$|\psi_1\rangle = U_{H1}(|\psi_0\rangle \otimes |\vec{0}\rangle) = \frac{1}{\sqrt{|C_2|}}\sum_{y\in C_2}(-1)^{(x+y)\cdot e_2}|x+y+e_1\rangle \otimes |H_1 e_1\rangle$$

Step 2: Measuring the ancilla to obtain the result $H_1 e_1$ and discarding the ancilla bits:

$$|\psi_2\rangle = \mathrm{tr}_{\mathcal{H}}|\psi_1\rangle = \frac{1}{\sqrt{|C_2|}}\sum_{y\in C_2}(-1)^{(x+y)\cdot e_2}|x+y+e_1\rangle$$

Step 3: Applying NOT gates to the qubits at whichever positions in the error e_1 a bit flip occurred:

$$|\psi_3\rangle = U_{e1}|\psi_2\rangle = \frac{1}{\sqrt{|C_2|}}\sum_{y\in C_2}(-1)^{(x+y)\cdot e_2}|x+y\rangle$$

Step 4: Using hadamard transform :

$$|\psi_4\rangle = H^{\otimes n}|\psi_3\rangle = \sqrt{\frac{|C_2|}{2^n}} \sum_{w\in C_2^\perp} (-1)^{x\cdot w}\,|w+e_2\rangle$$

Lemma 19

$$H^{\otimes n}|\psi_3\rangle = \sqrt{\frac{|C_2|}{2^n}} \sum_{w\in C_2^\perp} (-1)^{x\cdot w}\,|w+e_2\rangle$$

Proof For $|\psi_3\rangle = \frac{1}{\sqrt{|C_2|}} \sum_{y\in C_2} (-1)^{(x+y)\cdot e_2}\,|x+y\rangle$, we have

$$\begin{aligned}
H^{\otimes n}|\psi_3\rangle &= \frac{1}{\sqrt{|C_2|}} \sum_{y\in C_2} (-1)^{(x+y)\cdot e_2}\, H^{\otimes n}|x+y\rangle \\
&= \frac{1}{\sqrt{|C_2|}} \sum_{y\in C_2} (-1)^{(x+y)\cdot e_2}\, \frac{1}{\sqrt{2^n}} \sum_{z\in \mathbb{F}_2^n} (-1)^{(x+y)\cdot z}\,|z\rangle \\
&= \frac{1}{\sqrt{2^n|C_2|}} \sum_{z\in \mathbb{F}_2^n}\sum_{y\in C_2} (-1)^{(x+y)\cdot(e_2+z)}\,|z\rangle = \frac{1}{\sqrt{2^n|C_2|}} \sum_{w\in \mathbb{F}_2^n}\sum_{y\in C_2} (-1)^{(x+y)\cdot w}\,|w+e_2\rangle \\
&= \frac{1}{\sqrt{2^n|C_2|}} \sum_{w\in \mathbb{F}_2^n} (-1)^{x\cdot w} \left(\sum_{y\in C_2} (-1)^{y\cdot w}\right) |w+e_2\rangle \\
&= \frac{1}{\sqrt{2^n|C_2|}} \sum_{w\in C_2^\perp} (-1)^{x\cdot w} \left(\sum_{y\in C_2} (-1)^{y\cdot w}\right) |w+e_2\rangle \\
&= \frac{1}{\sqrt{2^n|C_2|}} \sum_{w\in C_2^\perp} (-1)^{x\cdot w}\,|C_2||w+e_2\rangle = \sqrt{\frac{|C_2|}{2^n}} \sum_{w\in C_2^\perp} (-1)^{x\cdot w}\,|w+e_2\rangle
\end{aligned}$$

■

Step 5: Introducing an ancilla containing sufficient qubits, and initially in the all zero state $|\vec{0}\rangle$.

$$|\psi_4\rangle \mapsto |\psi_4\rangle \otimes |\vec{0}\rangle$$

Using reversible computation to apply the parity matrix H_2 for the code $C_2^\perp$:

$$|\psi_5\rangle = U_{H2}\left(|\psi_4\rangle \otimes |\vec{0}\rangle\right) = \sqrt{\frac{|C_2|}{2^n}} \sum_{w\in C_2^\perp} (-1)^{x\cdot w}\,|w+e_2\rangle \otimes |H_2 e_2\rangle$$

Step 6: Measuring the ancilla to obtain the result $H_2 e_2$ and discarding the ancilla bits:

$$|\psi_6\rangle = \mathrm{tr}_{\mathcal{H}}|\psi_5\rangle = \sqrt{\frac{|C_2|}{2^n}} \sum_{w\in C_2^\perp} (-1)^{x\cdot w}\,|w+e_2\rangle$$

Step 7: Applying NOT gates to the qubits at whichever positions in the error e_2 a bit flip occurred:

$$|\psi_7\rangle = U_{e2}|\psi_6\rangle = \sqrt{\frac{|C_2|}{2^n}} \sum_{w\in C_2^\perp} (-1)^{x\cdot w}\,|w\rangle$$

Step 7: Using hadamard transform :

Lemma 20

$$H^{\otimes n}|\psi_7\rangle = |\psi\rangle.$$

Proof Using the lemma

$$H^{\otimes n}|\psi_7\rangle = \sqrt{\frac{|C_2|}{2^n}} \sum_{w \in C_2^{\perp}} (-1)^{x \cdot w}\, H^{\otimes n}|w\rangle = \sqrt{\frac{|C_2|}{2^n}} \sum_{w \in C_2^{\perp}} (-1)^{x \cdot w} \frac{1}{\sqrt{2^n}} \sum_{z \in \mathbb{F}_2^n} (-1)^{z \cdot w}\, |z\rangle$$
$$= \frac{\sqrt{|C_2|}}{2^n} \sum_{z \in \mathbb{F}_2^n} \sum_{w \in C_2^{\perp}} (-1)^{z \cdot w + x \cdot w}\, |z\rangle = \frac{\sqrt{|C_2|}}{2^n} \sum_{z \in \mathbb{F}_2^n} \sum_{w \in C_2^{\perp}} (-1)^{(z+x) \cdot w}\, |z\rangle$$

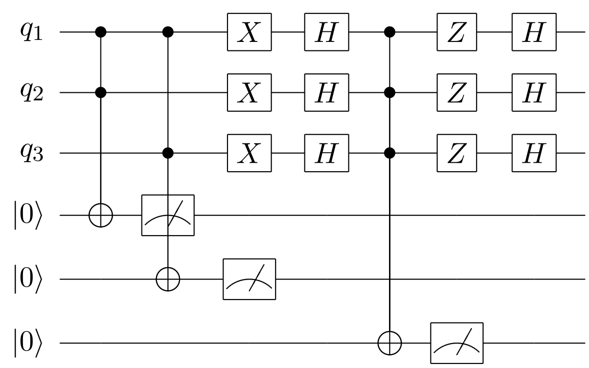
Since $z \mapsto z + x$ is an bijection of $\mathbb{F}_2^n$, we have

$$H^{\otimes n}|\psi_7\rangle = \frac{\sqrt{|C_2|}}{2^n} \sum_{z \in \mathbb{F}_2^n} \sum_{w \in C_2^{\perp}} (-1)^{z \cdot w}\, |z + x\rangle = \frac{\sqrt{|C_2|}}{2^n} \sum_{z \in C_2} \left(\sum_{w \in C_2^{\perp}} (-1)^{z \cdot w} \right) |z + x\rangle$$
$$= \frac{\sqrt{|C_2|}}{2^n} \sum_{z \in C_2} |C_2^{\perp}| |z + x\rangle = \frac{\sqrt{|C_2|}}{2^n} 2^{n - \dim C_2} \sum_{z \in C_2} |z + x\rangle$$
$$= \frac{1}{\sqrt{|C_2|}} \sum_{z \in C_2} |x + z\rangle = |\psi\rangle.$$

■

Quantum circuit of recover operation

For $n = 3$, $C_1 = [3, 1, 3]$ repetition code, $C_2 = [3, 0, 3]$ trivial code (satisfies $C_2 \subset C_1$), the complete recovery circuit is:



6.4 Plan for next week

1. TextBook chap 10.
2. The quantum circuit of CSS code.
3. Topological code, suface code.
4. Improve the categorification of errorcorrecting codes, quantum errorcorrection .

6.5 Appendix

categorification of errorcorrecting codes

Definition 21 A **general error-correction model** consists of:

- $\mathcal{M}$: message set,
- $\mathcal{X}$: codeword set,
- $\mathcal{E}$: set of error patterns, and for each $e \in \mathcal{E}$, a channel map $\tau_e \colon \mathcal{X} \to \mathcal{X}$,
- encoding map $\mathrm{Enc} \colon \mathcal{M} \to \mathcal{X}$.
- decoding map $\mathrm{Dec} \colon \mathcal{X} \to \mathcal{M}$.

The model is **error-correcting** if for all $e \in \mathcal{E}$, the diagram commutes:

$$\begin{tikzcd} \mathcal{M} \arrow[r, "\mathrm{Enc}"] \arrow[d, "\mathrm{Id}"'] & \mathcal{X} \arrow[d, "\tau_e"] \\ \mathcal{M} & \mathcal{X} \arrow[l, "\mathrm{Dec}"] \end{tikzcd}$$

Equivalently,

$$\mathrm{Dec} \circ \tau_e \circ \mathrm{Enc} = \mathrm{Id}_{\mathcal{M}}.$$

Such an error-correcting model is called an **object** of the category **EC**. And we denote an object by $\mathcal{C} = (\mathcal{M}, \mathcal{X}, \mathrm{Enc}, \mathrm{Dec}, \{\tau_e\}_{e\in\mathcal{E}})$.

Definition 22 Let

$$\mathcal{C}_1 = (\mathcal{M}_1, \mathcal{X}_1, \mathrm{Enc}_1, \mathrm{Dec}_1, \{\tau_e\}_{e\in\mathcal{E}_1}), \quad \mathcal{C}_2 = (\mathcal{M}_2, \mathcal{X}_2, \mathrm{Enc}_2, \mathrm{Dec}_2, \{\sigma_f\}_{f\in\mathcal{E}_2})$$

be two objects of **EC**. A **morphism** from $\mathcal{C}_1$ to $\mathcal{C}_2$ is a triple

$$F = (f,\ g,\ \varphi)$$

where

- $f : \mathcal{M}_1 \to \mathcal{M}_2$,
- $g : \mathcal{X}_1 \to \mathcal{X}_2$,
- $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$,

satisfying the following three commutativity conditions:

1. **Encoding preservation**: $g \circ \mathrm{Enc}_1 = \mathrm{Enc}_2 \circ f$;
2. **Decoding preservation**: $f \circ \mathrm{Dec}_1 = \mathrm{Dec}_2 \circ g$;
3. **Error structure preservation**: $\forall e \in \mathcal{E}_1,\ g \circ \tau_e = \sigma_{\varphi(e)} \circ g$.

Composition of morphisms is defined componentwise:

$$(f_2, g_2, \varphi_2) \circ (f_1, g_1, \varphi_1) := (f_2 \circ f_1,\ g_2 \circ g_1,\ \varphi_2 \circ \varphi_1).$$

The identity morphism on an object $\mathcal{C} = (\mathcal{M}, \mathcal{X}, \mathrm{Enc}, \mathrm{Dec}, \{\tau_e\}_{e\in\mathcal{E}})$ is

$$\mathrm{id}_{\mathcal{C}} := (\mathrm{id}_{\mathcal{M}},\ \mathrm{id}_{\mathcal{X}},\ \mathrm{id}_{\mathcal{E}}).$$

The morphism can be drawed as a commutative diagram :

$$
\begin{tikzcd}
M_1 \arrow[r, "\mathrm{Enc}_1"'] \arrow[rr, bend left, "f"] \arrow[d, "1_{M_1}"'] & X_1 \arrow[rr, bend left, "g"] \arrow[d, "\tau_e"] & M_2 \arrow[r, "\mathrm{Enc}_2"'] \arrow[d, "1_{M_2}"'] & X_2 \arrow[d, "\sigma_{\varphi_1(e)}"] \\
M_1 \arrow[rr, bend right, "f"'] & X_1 \arrow[l, "\mathrm{Dec}_1"'] \arrow[rr, bend right, "g"'] & M_2 & X_2 \arrow[l, "\mathrm{Dec}_2"']
\end{tikzcd}
$$

we can use this diagram to prove composition of morphism is a morphism:

$$
\begin{tikzcd}
M_1 \arrow[r, "\mathrm{Enc}_1"'] \arrow[rr, bend left, "f_1"] \arrow[d, "1_{M_1}"'] & X_1 \arrow[rr, bend left, "g_1"] \arrow[d, "\tau_e"] & M_2 \arrow[r, "\mathrm{Enc}_2"'] \arrow[rr, bend left, "f_2"] \arrow[d, "1_{M_2}"'] & X_2 \arrow[rr, bend left, "g_2"] \arrow[d, "\tau_{\varphi_1(e)}"] & M_3 \arrow[r, "\mathrm{Enc}_3"'] \arrow[d, "1_{M_3}"'] & X_3 \arrow[d, "\tau_{\varphi_2 \circ \varphi_1(e)}"] \\
M_1 \arrow[rr, bend right, "f_1"'] & X_1 \arrow[l, "\mathrm{Dec}_1"'] \arrow[rr, bend right, "g_1"'] & M_2 \arrow[rr, bend right, "f_2"'] & X_2 \arrow[l, "\mathrm{Dec}_2"'] \arrow[rr, bend right, "g_2"'] & M_3 & X_3 \arrow[l, "\mathrm{Dec}_3"']
\end{tikzcd}
$$

Some ideas of categorification of quantum errorcorrection models

Lemma 23 *Let $\mathcal{H}_1, \mathcal{H}_2$ be finite-dimensional complex Hilbert spaces, and let $A\colon \mathcal{H}_1 \to \mathcal{H}_2$ be injective linear. Define $\mathbb{P}\mathcal{H} := \big(\mathcal{H} \setminus \{0\}\big)/\mathbb{C}^*$. Then A induces a well-defined map $\widetilde{A}\colon \mathbb{P}\mathcal{H}_1 \to \mathcal{L}(\mathcal{H}_2)$ by*

$$\widetilde{A}\big([\psi]\big) := \frac{A|\psi\rangle\big(A|\psi\rangle\big)^\dagger}{\langle\psi \mid \psi\rangle}.$$

Definition 24 A **quantum errorcorrection model** consists of the primitive data:

- $\mathcal{H}_{\mathrm{L}}, \mathcal{H}_{\mathrm{P}}$: finitedimensional complex Hilbert spaces (logical space, physical space),
- $\mathcal{E}\colon \mathcal{L}(\mathcal{H}_{\mathrm{P}}) \to \mathcal{L}(\mathcal{H}_{\mathrm{P}})$: quantum operation (error channel),
- $\mathrm{Enc}\colon \mathcal{H}_{\mathrm{L}} \to \mathcal{H}_{\mathrm{P}}$: linear encoding map,
- $\mathrm{Dec}\colon \mathcal{L}(\mathcal{H}_{\mathrm{P}}) \to \mathbb{P}(\mathcal{H}_{\mathrm{L}})$: decoding map.

The following are **derived structures**, not part of the primitive definition:

- $\mathbb{P}(\mathcal{H}_{\mathrm{L}})$: projective space of rays in $\mathcal{H}_{\mathrm{L}}$,
- $\mathcal{L}(\mathcal{H}_{\mathrm{P}})$: space of bounded linear operators on $\mathcal{H}_{\mathrm{P}}$,
- $\widetilde{\mathrm{Enc}}\colon \mathbb{P}(\mathcal{H}_{\mathrm{L}}) \to \mathcal{L}(\mathcal{H}_{\mathrm{P}})$: induced encoding on projective space, canonically induced by Enc.

The model is **errorcorrecting** if

$$\mathrm{Dec} \circ \mathcal{E} \circ \widetilde{\mathrm{Enc}} = \mathrm{Id}_{\mathbb{P}(\mathcal{H}_{\mathrm{L}})}.$$

Equivalently, the diagram commutes:

$$\begin{tikzcd}
\mathbb{P}(\mathcal{H}_{\mathrm{L}}) \arrow[r, "\widetilde{\mathrm{Enc}}"] \arrow[d, "\mathrm{Id}"'] & \mathcal{L}(\mathcal{H}_{\mathrm{P}}) \arrow[d, "\mathcal{E}"] \\
\mathbb{P}(\mathcal{H}_{\mathrm{L}}) & \mathcal{L}(\mathcal{H}_{\mathrm{P}}) \arrow[l, "\mathrm{Dec}"']
\end{tikzcd}$$

Definition 25 Let

$$\big(\mathcal{H}_{\mathrm{L},1}, \mathcal{H}_{\mathrm{P},1}, \mathcal{E}_1, \mathrm{Enc}_1, \mathrm{Dec}_1\big), \quad \big(\mathcal{H}_{\mathrm{L},2}, \mathcal{H}_{\mathrm{P},2}, \mathcal{E}_2, \mathrm{Enc}_2, \mathrm{Dec}_2\big)$$

be two quantum errorcorrection models (primitive data only). Let

$$\mathbb{P}(\mathcal{H}_{\mathrm{L},i}),\ \mathcal{L}(\mathcal{H}_{\mathrm{P},i}),\ \widetilde{\mathrm{Enc}}_i$$

denote the derived projective space, operator space, and induced encoding for each model $i = 1, 2$.
A **morphism** between them is a pair of maps

$$F\colon \mathbb{P}(\mathcal{H}_{\mathrm{L},1}) \to \mathbb{P}(\mathcal{H}_{\mathrm{L},2}), \quad G\colon \mathcal{L}(\mathcal{H}_{\mathrm{P},1}) \to \mathcal{L}(\mathcal{H}_{\mathrm{P},2}),$$

satisfying:

1. Commute with induced encoding:
$$G \circ \widetilde{\mathrm{Enc}}_1 = \widetilde{\mathrm{Enc}}_2 \circ F,$$
2. Commute with decoding:
$$F \circ \mathrm{Dec}_1 = \mathrm{Dec}_2 \circ G,$$
3. Commute with quantum error operation:
$$G \circ \mathcal{E}_1 = \mathcal{E}_2 \circ G.$$

References

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